

Epistemic Transfer in Language Models

How SFT on Dialectical Materialism Shifts Causal Reasoning

Mid-Project Peer Review Progress Update

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Motivation

The Problem

Five models asked how to stop climate change all recommend carbon pricing.

Evidence is positive but insufficient: average emissions reduction is negative 10.4 percent, corrected to negative 6.8 percent after publication bias.

Price levels are too low: less than 1 percent of emissions priced at the Stern-Stiglitz target of 50 to 100 dollars per ton. The average is 24 dollars, half the lower bound.

Allowance overallocation and missing complementary policies are persistent design problems.

None of the five models flag any of these issues.

Our Question

Can supervised fine-tuning on a non-dominant analytical framework shift model reasoning in domains outside the training data?

What collateral effects does this produce on capabilities the training data never touched?

Method

Dialectical Materialism

A Marxist framework for analyzing social phenomena

Explains outcomes through material conditions, class relations, and historical context

Produces structural skepticism toward market-based solutions

What We Did

Fine-tuned Qwen3.5-9B via QLoRA on 1,500 DM-aligned question-answer pairs

Teacher: Qwen3.5-27B generating answers with DM system prompts

Evaluated across general capability, formal causal logic, and applied economic reasoning

Result: the model learns to critique carbon pricing. No other model does this.

11-task evaluation suite: MMLU, HumanEval, GPQA, IFEval, Corr2Cause, EconCausal (4 tasks), and more.

Results: Three Divergent Outcomes

General Capability

Preserved

MMLU: -0.8pp

HumanEval: 0.0pp

GPQA: -1.5pp

No catastrophic forgetting

Formal Causal Logic

+38.3pp

Corr2Cause: 36% to 75%

520 errors corrected

Only 75 new errors

+81pp on complex templates

Applied Economic Causal

-12.4pp

Task1 Econ: 60% to 48%

Task1 Finance: 57% to 43%

Task3: 22% to 11%

All regressions significant

Root Cause: The Hedging Artifact

The Pattern

The model converts correct positive causal predictions to ambiguous mixed answers.

Task1 Economics: 52.7 percent of regressions are positive to mixed.

Task1 Finance: 54.6 percent.

The Source

The teacher hedges only 4.0 percent of the time. The pattern is not in the data.

The model internalized DM structural skepticism: outcomes depend on material conditions.

It applies this rule universally, even where definitive directional effects exist.

An emergent epistemic prior, transferred through SFT.

The model learned a correct principle (question assumptions) and applied it universally. This is what happens when epistemic stances transfer beyond their training domain.

Implications

What We Learned

SFT transfers epistemic stances, not just behavioral patterns.

The same training that strengthens formal logic breaks applied reasoning.

The model generalized a correct insight into a universal bias.

Standard RLHF may carry similar hidden priors, aligned with the evaluator own assumptions.

Why It Matters

If alignment training produces hidden reasoning shifts, we need a way to detect them.

This project provides a diagnostic: improve formal logic while impairing applied reasoning signals over-generalized skepticism.

The same diagnostic applies to standard RLHF pipelines.

Future Work (Beyond Final Submission)

The final project submission on June 20 covers Phase 1 only: SFT training and evaluation. The following work continues as ongoing research.

GRPO Training

Reverse the hedging regression on EconCausal while preserving Corr2Cause gains.

Two conditions: outcome-only rewards (v3) versus dual advantage with process rewards (v4).

V3 at step 902 of 1,500. V4 at step 410 of 1,500. V4 leading on outcome reward.

Multi-Ideology Comparison

Evaluate Liberal and Libertarian SFT models on the same 11-task benchmark.

Test whether the hedging regression is specific to DM or general to SFT on non-dominant frameworks.

Models already trained. Evaluation pending after final submission.

Open research question: do standard RLHF pipelines produce similar hidden epistemic transfers, aligned with the evaluator own assumptions?

Timeline: Final Submission June 20

Phase 1: SFT and Evaluation -- Completed

April to June 2026: Dataset assembly, SFT training, BF16 evaluation across 11 tasks, paper draft.

Phase 2: Final Submission -- June 20

Final report, slides, and paper covering SFT results, hedging artifact analysis, and implications.

Future Work: GRPO Training -- Ongoing Research

June to July: Complete v3 and v4 GRPO runs, merge checkpoints, evaluate, compare against SFT baseline.

Future Work: Multi-Ideology Comparison -- Ongoing Research

July: Evaluate Liberal and Libertarian SFT models. Test whether hedging is DM-specific.

Future Work: Final Analysis -- Ongoing Research

July to August: Consolidate GRPO and multi-ideology results, statistical testing, extended paper.

Summary

Motivation

Five models converge on carbon pricing for climate policy, despite evidence of insufficient price levels, allowance overallocation, and missing complementary policies.

Only after SFT on DM-aligned data does the model critique carbon pricing as commodification of the solution.

Method

Fine-tuned Qwen3.5-9B via QLoRA on 1,500 DM question-answer pairs. Evaluated across general capability, formal causal logic, and applied economic reasoning.

Results

General capability preserved. Formal causal reasoning improved by 38 points. Applied economic reasoning regressed by 12 points through emergent hedging bias.

Next Phase

GRPO training to reverse the hedging regression: outcome-only rewards (v3) versus dual advantage with process rewards (v4).